Marcio Mastrocola Alcantara

Software Architect · Tech Lead

Londrina, Brazil Remote only CLT, PJ or contractor Available in 30 days

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Software architect with twenty years building systems that cannot stop, mostly in fintech. Led the migration of a credit cooperative's backend from a synchronous monolith to event-driven microservices: process execution time down about 50% and feature lead time from four sprints to two. Now putting AI into production on the same terms — LLM agents with budgets, guardrails and audit trails — built in public at mastrocola.dev.

EXPERIENCE

Sisprime do Brasil *Software Engineer, backend architecture lead* Aug 2023 – present

- Led the migration of the business backend from a legacy synchronous monolith to event-driven microservices, without interrupting operations
- Cut process execution time by about 50% and halved feature lead time, from four sprints to two

Node.js, Python, Airflow, SQL Server, RabbitMQ, Elasticsearch, Docker, Linux

Proffer *Data Engineer* Apr 2023 – Aug 2023

- Owned the ERP integration strategy for the client base, from technical discovery with stakeholders to production
- Replaced fragile, manual processes with automated pipelines, removing a recurring source of multi-day delays; also served as the company's DBA

Apache Airflow, Python, AWS, SQL

Mutual *Data Engineer · Software Engineer* Nov 2021 – Feb 2023

- Designed the company's data platform from scratch: ETL architecture, batch and streaming pipelines centralizing data from the microservices platform, reporting decoupled from production and real-time data across the company
- Technical lead of the data team: sprint planning, roadmap and agile ceremonies
- Before that, designed the credit decision environment of a solar financing platform on BPMN and DMN, making the decision engine fully parametrizable and cutting change turnaround from over an hour to minutes; led the five-person team behind credit and collection workflows

Python, Node.js, AWS (Lambda, Glue, Athena, Kinesis, ECS, S3, SQS, SNS), Airflow, MongoDB, Terraform, Docker, Grafana, Kibana

Uniprime Central *Software Engineer* Sep 2015 – Apr 2021

- Developed and maintained the credit cooperative's core system and its third-party integrations; owned integration and deployment routines for two years
- Optimized the critical SQL processes behind regulatory reporting, with substantial gains in the generation of compliance files

PowerBuilder, C#, SQL Server

Orium Software *Software Engineer* Jan 2012 – Jun 2015

- Built core ERP modules (finance, production planning, purchasing, fiscal documents) and the point-of-sale system with fiscal printers, pinpads and label printers
- Owned fiscal certification end to end: PAF-ECF, card payment terminals and NFC-e electronic invoices with the certifying bodies

Delphi, Oracle

Guenka Software *Software Engineer* May 2007 – May 2011

- Embedded C for vehicle tracking devices; later coordinated a five-person QA team and built test automation for vehicle comfort modules for national and Japanese automakers

Veltec Soluções Tecnológicas *Software Engineer, intern before* Aug 2005 – May 2007

- Logistics and fleet management systems with vector map tracking and client-server communication over raw TCP/UDP sockets

BUILT IN PUBLIC

mastrocola-dev *Architecture portfolio run as a company* 2026

- LLM agent runtime in production on Azure: queue-based jobs on Functions, per-visitor quotas, daily spend cap, prompt-injection rules, tools only through MCP, traces
- Zero secrets in GitHub: one workload identity per repository over OIDC, Key Vault as the only store; every decision recorded as an ADR
- Household finance ledger with a Claude connector: every write audited, nothing deleted, joint decisions confirmed by both members, React web app

TypeScript, Terraform, Azure, Cloudflare, PostgreSQL, Anthropic API, Model Context Protocol

SKILLS

Architecture	Event-driven microservices, legacy modernization, BPMN/DMN, ADRs
Backend	Node.js and TypeScript, Python; Express, Nest.js, FastAPI, Flask
Data	SQL Server, PostgreSQL, Oracle, MongoDB, DynamoDB; Airflow, PySpark, Glue, Athena
Messaging	RabbitMQ, Kafka, Kinesis, Azure Service Bus, Redis
Cloud and DevOps	AWS, Azure, Cloudflare, Terraform, Docker, Kubernetes, GitHub Actions
AI	LLM agents, Model Context Protocol, Anthropic API
Observability	Grafana, Elasticsearch and Kibana, CloudWatch

EDUCATION AND LANGUAGES

2019 – 2021	Postgraduate studies in Big Data — SENAI
2015 – 2019	Civil Engineering — Universidade Norte do Paraná
2002 – 2005	Computer Science — Universidade Estadual de Londrina
Certification	CPA-10 (ANBIMA)
Languages	Portuguese (native), English (full professional)